Supplementary Materials: Chill-Window Attribution of ENSO-Driven Olive Yield Collapse on the Hyper-Arid Peruvian Coast

Javier Quille-Mamani [1], José Huanuqueño-Murillo [2], David Quispe-Tito [2], German Huayna-Felipe [3], Jorge Espinoza-Molina [4], Karina Acosta-Caipa [4], Heler Samir Pérez-Cubas [5], Eusebio Ingol-Blanco [6], Lia Ramos-Fernández [2,*] and Edwin Pino-Vargas [7,*]

Supplementary Figures

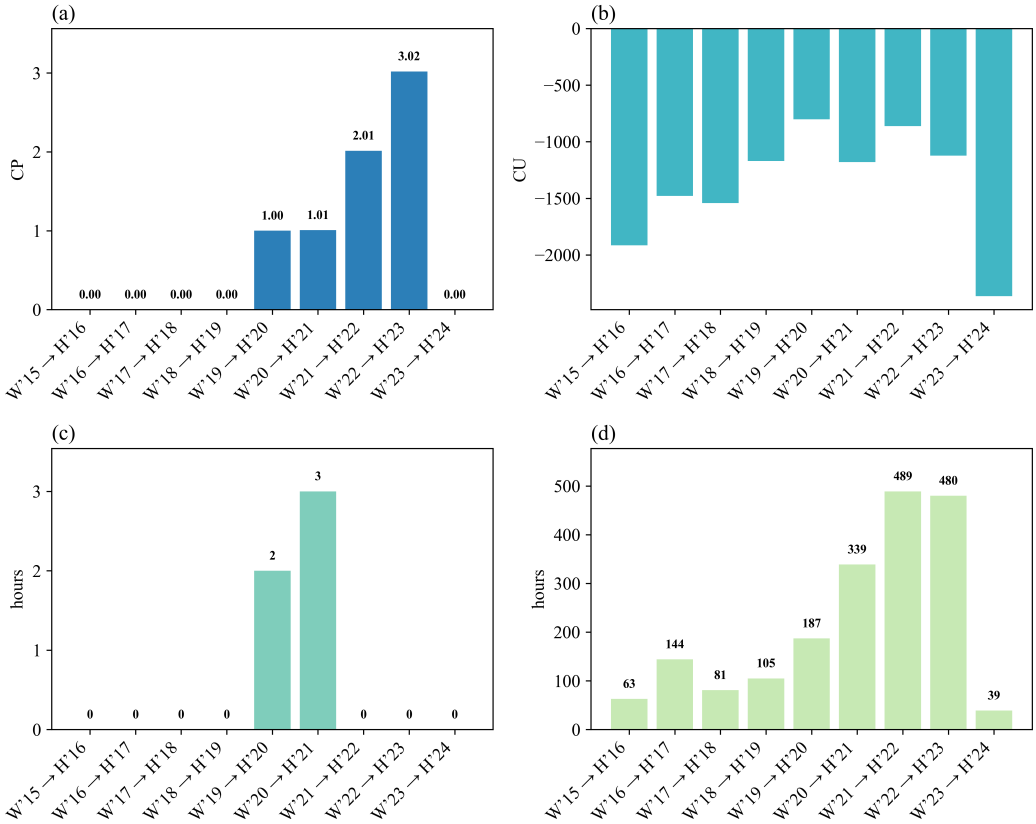

Figure S1. Inter-comparison of chill-accumulation metrics over the May–August chill window ($Y-1$) for the 2016–2024 study period. (**a**) Dynamic-Model Chill Portions (CP); (**b**) Utah Chill Units (CU); (**c**) hours below 7.2 °C; (**d**) hours below 12 °C. Bars are labelled "W′YY → H′YY+1" (chill winter → harvest year). CP floors at zero in five of eight years, motivating the use of the chill-window mean temperature $T_{\text{chill},t-1}$ as the primary chill predictor.

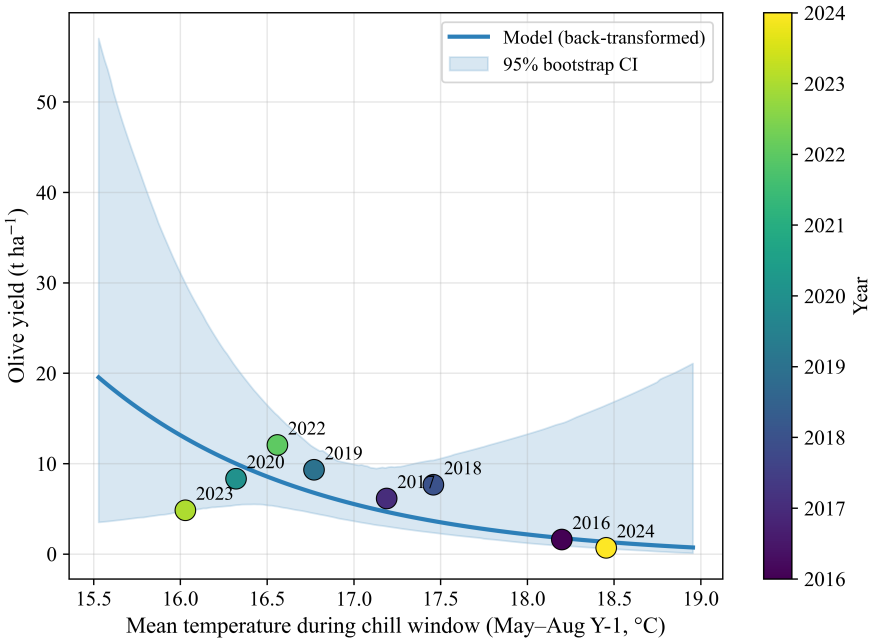

Figure S2. Partial-effect plot of $T_{\text{chill},t-1}$ on back-transformed olive yield (t ha^{-1}), holding $T_{\text{CREC},t}$ at its observed mean. Solid line: primary log-OLS prediction back-transformed to the raw t ha^{-1} scale; shaded band: year-block bootstrap 95 % CI ($B = 5{,}000$). Points are observed yields colour-coded by harvest year. The negative log-space slope ($\beta_{\text{chill}} = -0.816$, $p = 0.030$) translates into the convex curve shown here.

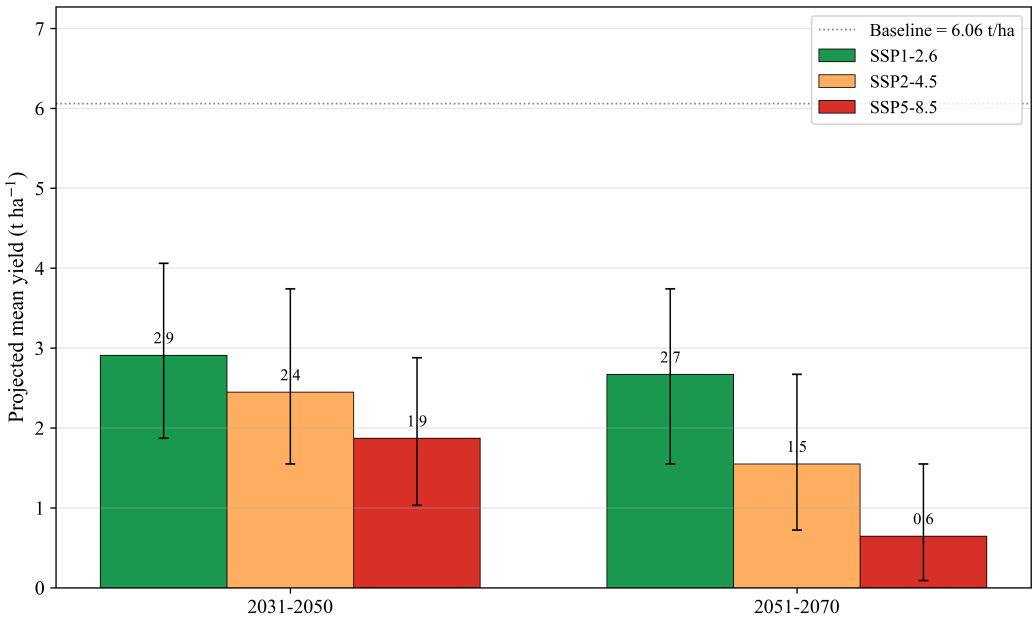

Figure S3. CMIP6 delta-method projections of mean olive yield under three SSP pathways (SSP1-2.6, SSP2-4.5, SSP5-8.5) and two time horizons (2031–2050, 2051–2070), relative to the calibrated baseline yield ($\bar{y}_0 = 6.06$ t ha^{-1}; dashed line). Bars are central estimates obtained with the IPCC AR6 WGI Atlas median warming for region SWS; whiskers span the 5–95 % inter-model envelope (Supplementary Table S3). Horizons beyond 2070 are omitted as extrapolative.

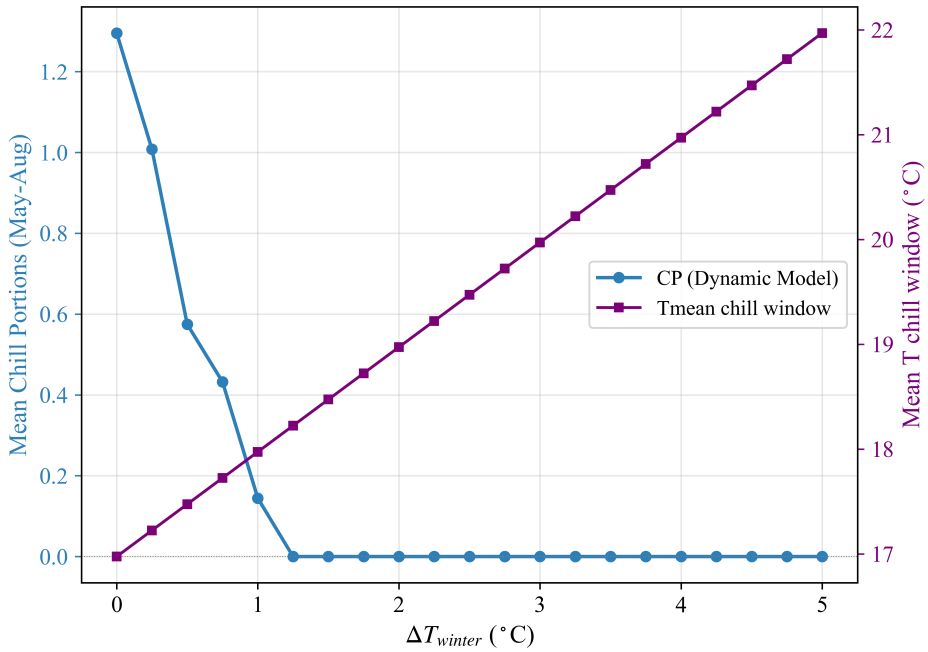

Figure S4. Chill-collapse mechanism: Dynamic-Model Chill Portions (CP, left axis, blue) and chill-window mean temperature T_{chill} (right axis, purple) as functions of a uniform winter warming ΔT_{winter} applied to the 2015–2025 chill-window daily series. CP collapses to zero at $\Delta T_{\text{winter}} \approx +1.25\,^{\circ}\text{C}$, whereas T_{chill} rises continuously, motivating T_{chill} as the chill predictor in the warm tail.

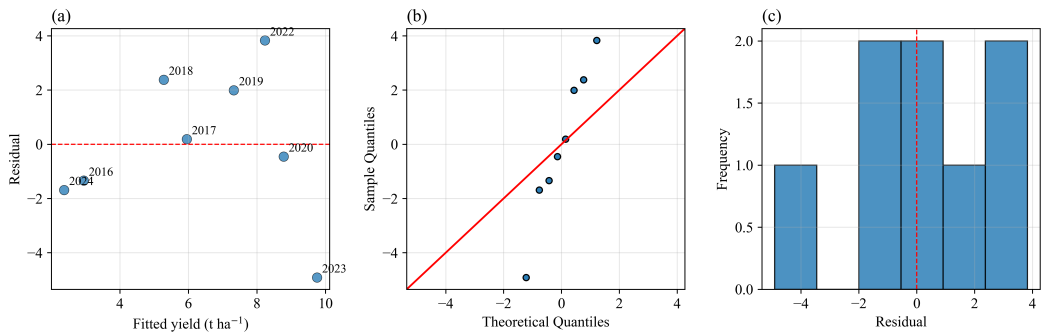

Figure S5. In-sample residual diagnostics for the primary log-OLS model on the back-transformed raw-yield scale (t ha^{-1}). (**a**) Residuals vs. fitted yield, labelled by harvest year; (**b**) Normal Q–Q plot; (**c**) Histogram of residuals. Residuals are approximately Gaussian and homoscedastic within the limits of $n = 8$; leverage and Cook's-distance diagnostics are discussed in Section 4.5 of the main text.

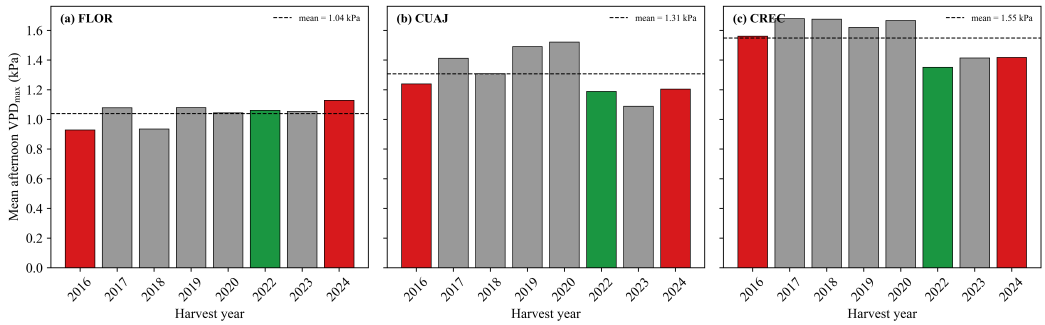

Figure S6. Afternoon $\mathrm{VPD}_{\max}$ proxy ($e_s(T_{\max})\,(1-\mathrm{RH}_{\min}/100)$) averaged within the FLOR, CUAJ and CREC phenology windows of each harvest year. Red: 2016 and 2024 failure years; green: 2022 bumper year; dashed line: eight-year mean. The two failure years lie within $\pm 10\,\%$ of the eight-year mean, ruling out atmospheric dryness as an alternative driver of the observed yield collapses (Section 4.5 of the main text).

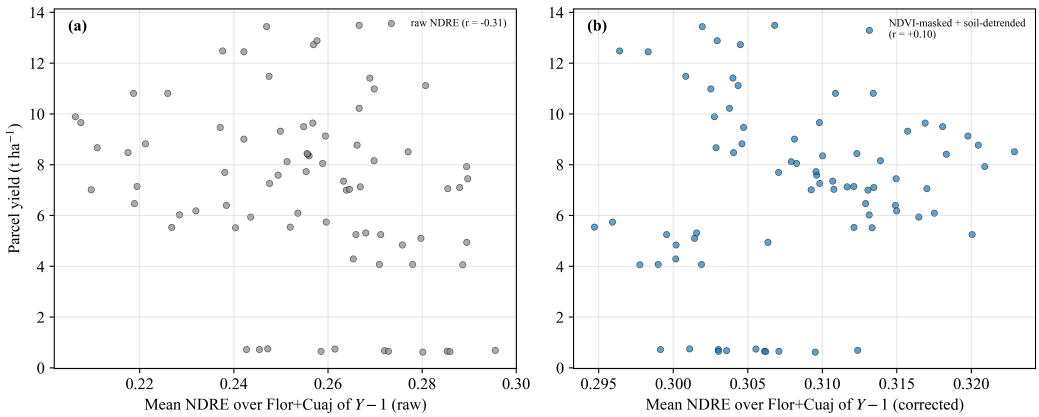

Figure S7. Mixed-pixel mitigation for Sentinel-2 NDRE over the open-canopy La Yarada parcels ($n = 77$ parcel-years). (**a**) Raw parcel-mean NDRE composited over Flor+Cuaj of $Y-1$ vs. parcel-level annual yield. (**b**) After an NDVI > 0.25 bare-soil mask and first-order linear soil-detrending referenced to the canopy endmember $\mathrm{NDVI}_{\mathrm{pure}} = 0.42$. The pooled Pearson correlation rises from $r = -0.31$ (raw) to $r = +0.10$ (corrected), motivating the relegation of NDRE to a parcel-level descriptive role (Section 3.6 of the main text).

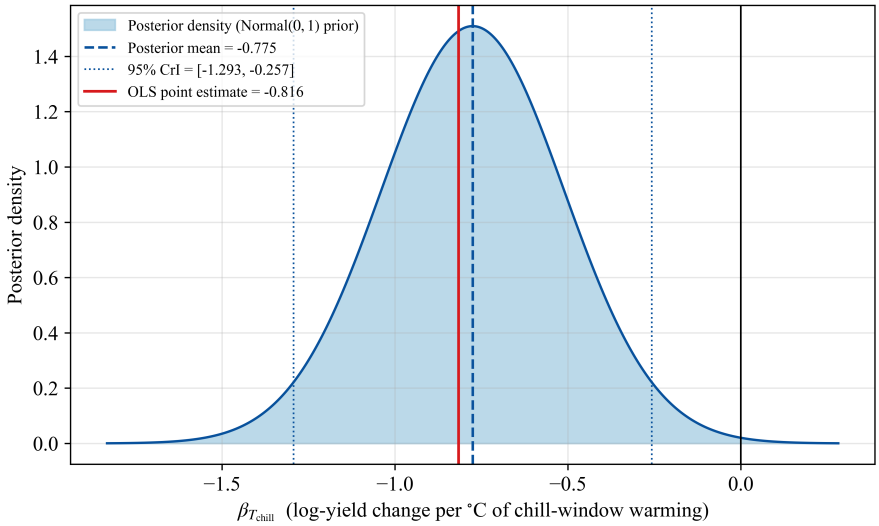

Figure S8. Closed-form Gaussian posterior of the chill slope $\beta_{T_{\text{chill}}}$ (natural scale) under a weakly-informative Normal$(0, 1)$ prior on the standardised coefficient. Posterior mean -0.775, 95 % CrI $[-1.293, -0.257]$; OLS point estimate -0.816 (red) is indistinguishable from the posterior mean. The posterior mass lies entirely below zero (Savage–Dickey $\text{BF}_{10} = 15.9$), corroborating the frequentist evidence (Section 3.3 of the main text).

Supplementary Tables

Table S1. Robustness summary of the primary attribution model across alternative predictor pairs and cross-validation strategies. Metrics on the raw t ha^{-1} scale except *loyo_log* rows (log-space). Variants: *in_sample* = full-calibration fit; *loyo_raw* = unbounded raw LOYO; *loyo_clipped* = raw LOYO clipped to $[0, 1.1 \cdot y_{\max}]$; *loyo_log* = log-link LOYO; *loyo_interior* = LOYO restricted to interior years (excluding the 2016 and 2024 warm-tail folds).

Variant	Stage	n	R^2	RMSE	MAE	Bias
Primary: T_{chill} + T_{CREC}						
	in_sample	8	+0.108	3.41	2.54	−0.40
	loyo_raw	8	**−1.694**	**5.92**	**4.26**	+0.31
	loyo_clipped	8	−0.756	4.78	3.69	−0.26
	loyo_log	8	−0.071	0.79	0.65	+0.08
	loyo_interior	4	+0.218	2.67	2.38	−0.96
T_{chill} *only*						
	in_sample	8	+0.104	3.41	2.70	−0.40
	loyo_raw	8	−1.452	5.65	4.00	+0.47
	loyo_clipped	8	−0.290	4.10	3.32	−0.21
	loyo_log	8	+0.199	0.69	0.58	+0.09
	loyo_interior	5	+0.079	2.60	2.22	−1.43
CP + T_{CREC}						
	in_sample	8	−0.174	3.91	3.68	−1.17
	loyo_raw	8	−15.608	14.70	10.11	+3.87
CH_{12} + T_{CREC}						
	in_sample	8	−0.175	3.91	3.55	−1.00
	loyo_raw	8	−8.648	11.20	8.25	+2.44
T_{chill} + $T_{\max,FLOR}$						
	in_sample	8	+0.230	3.16	2.55	−0.35
	loyo_raw	8	−1.139	5.27	4.12	+0.24
RF (5 features)						
	loyo_raw	8	−0.040	3.68	3.23	+0.41
$ONI_{May–Aug,t-1}$ + T_{CREC} *(ENSO robustness)*						
	in_sample	8	−0.080	3.75	3.19	—
	loyo_raw	8	−3.562	7.70	5.97	—
T_{chill} + T_{CREC} + *share_ON (alternance sensitivity)*						
	in_sample ($\beta_{T_{\text{chill}}} = -0.601$, $p = 0.031$)	8	+0.893	—	—	—

Table S2. Year-level Pearson correlations between mean olive yield and candidate climate predictors ($n = 8$ years). $T_{\text{chill},t-1}$ ($r = -0.697$) is the strongest single predictor; $T_{\text{CREC},t}$ was selected as the heat-side predictor for its low collinearity with $T_{\text{chill},t-1}$ ($r = 0.409$, VIF < 2) and its clean physiological decomposition. Threshold-based chill counts (CP, CH_{12}) are weaker because they saturate at zero in the warm tail.

Predictor	Pearson r vs yield	Note
$T_{\text{chill},t-1}$ (May–Aug, °C)	**−0.697**	primary (chill side)
$T_{\text{max,FLOR}}$ (°C)	−0.492	flowering $T_{\max}$
$CH_{12,t-1}$ (h)	+0.404	threshold chill, h< 12 °C
$T_{\text{mean,FLOR}}$ (°C)	−0.381	flowering window
$T_{\text{max,CREC}}$ (°C)	−0.375	fruit growth $T_{\max}$
$T_{\text{CREC},t}$ (Jan–Mar, °C)	**−0.313**	primary (heat side)
CP_{t-1} (Dynamic Model)	+0.307	saturated at 0 in 5/8 years
$ET_{0,\text{CREC}}$ (mm)	−0.178	FAO 56 reference ET, fruit growth
Cross-correlation $T_{chill,t-1} \times T_{CREC,t} = +0.409$ *(moderate; VIF* < 2*)*		

Table S3. CMIP6 warming envelope used for the delta-method projections: ensemble central estimates and 5–95 % inter-model spread from the IPCC AR6 WGI Atlas, region South-Western South America (SWS), relative to the 1995–2014 baseline. Horizons are restricted to 2031–2050 and 2051–2070; the SSP5-8.5 2071–2100 ensemble median (+4 °C) is omitted as a 3× extrapolation beyond the warmest observed chill year ($\Delta T_{\text{winter}} \approx +1.48\,^{\circ}\text{C}$).

Scenario	**Period**	$\Delta T_{\textbf{winter}}$ (°C)			$\Delta T_{\textbf{summer}}$ (°C)		
		5 %	median	95 %	5 %	median	95 %
SSP1-2.6	2031–2050	0.5	0.9	1.4	0.5	0.9	1.4
SSP1-2.6	2051–2070	0.6	1.0	1.6	0.6	1.0	1.6
SSP2-4.5	2031–2050	0.6	1.1	1.6	0.6	1.1	1.6
SSP2-4.5	2051–2070	1.0	1.6	2.3	1.0	1.6	2.2
SSP5-8.5	2031–2050	0.9	1.4	2.0	0.8	1.4	2.0
SSP5-8.5	2051–2070	1.6	2.4	3.3	1.6	2.4	3.2

Table S4. Phenology windows used to compute the climate and Sentinel-2 aggregates. The chill window (Reposo+BROT, 1 May–31 Aug of $Y-1$) is the canonical austral-winter chill window; floral induction (FLOR, CUAJ) occurs in $Y-1$; fruit growth (CREC) and maturation (ENV) occur in Y.

Window	**Start**	**End**	**Calendar year**
Reposo + BROT (chill window)	1 May	31 Aug	$Y-1$
BROT (budburst)	15 Aug	30 Sep	$Y-1$
FLOR (flowering)	1 Oct	15 Nov	$Y-1$
CUAJ (fruit set)	16 Nov	31 Dec	$Y-1$
CREC (fruit growth)	1 Jan	15 Mar	Y
ENV (on-tree maturation)	16 Mar	30 Apr	Y

Table S5. Vapour-pressure deficit (VPD, FAO 56) over the FLOR, CUAJ and CREC phenology windows for each calibration year, plus year-level Pearson correlations between VPD aggregates and observed yield ($n = 8$). VPD_{m} = within-window mean of daily mean VPD; $\text{VPD}_{\text{max,m}}$ = within-window mean of the daily afternoon proxy $e_s(T_{\max})\,(1 - \text{RH}_{\min}/100)$; $\text{VPD}_{\text{max},p95}$ = 95th percentile of the same proxy. 2016 and 2024 failure years (bold) are not VPD-anomalous relative to the 2022 bumper year (italic); yield–VPD correlations are weak ($|r| < 0.36$ for the mean and afternoon-mean metrics).

Year	**FLOR**			**CUAJ**			**CREC**		
	VPD_{m} (kPa)	$\text{VPD}_{\text{max,m}}$ (kPa)	$\text{VPD}_{\text{max},p95}$ (kPa)	VPD_{m} (kPa)	$\text{VPD}_{\text{max,m}}$ (kPa)	$\text{VPD}_{\text{max},p95}$ (kPa)	VPD_{m} (kPa)	$\text{VPD}_{\text{max,m}}$ (kPa)	$\text{VPD}_{\text{max},p95}$ (kPa)
2016	0.479	0.929	1.131	0.643	1.240	1.413	0.783	1.561	1.851
2017	0.581	1.079	1.238	0.637	1.412	1.662	0.845	1.679	2.130
2018	0.486	0.935	1.095	0.615	1.309	1.638	0.823	1.676	2.013
2019	0.549	1.079	1.290	0.724	1.491	1.761	0.752	1.620	1.963
2020	0.448	1.044	1.593	0.649	1.521	1.879	0.705	1.667	2.205
2022	*0.448*	*1.061*	*1.775*	*0.525*	*1.188*	*1.903*	*0.644*	*1.351*	*1.771*
2023	0.474	1.054	1.596	0.471	1.089	1.678	0.645	1.414	1.992
2024	0.491	1.128	1.586	0.547	1.204	1.679	0.642	1.418	1.898
Mean	0.495	1.039	1.413	0.601	1.307	1.702	0.730	1.553	1.978
r vs. yield	−0.08	+0.05	+0.26	+0.15	+0.36	+0.76	−0.00	+0.09	+0.07

Table S6. Leave-out sensitivity of the primary log-OLS model $\log(y + 0.5) \sim T_{\text{chill},t-1} + T_{\text{CREC},t}$, refit on the full eight-year sample, with 2024 removed, and with both warm-tail years (2016, 2024) removed. Removing 2024 alone halves the chill slope ($p = 0.37$); removing both warm-tail anchors collapses the chill-window range to ∼1.5 °C and flips the fitted slope (+0.32, $p = 0.44$), quantifying how much attribution skill is carried by the two extreme years.

Variant	n	$\beta_{T_{\text{chill}}}$ **(SE)**	p_{chill}	$\beta_{T_{\text{CREC}}}$	p_{CREC}	R^2_{log}	R^2_{raw}
Full (2016–2024, ex 2021)	8	−0.816 (0.272)	0.0299	+0.089	0.529	0.653	0.108
Leave 2024	7	−0.433 (0.428)	0.3681	−0.033	0.853	0.396	0.091
Leave {2016, 2024}	6	+0.322 (0.360)	0.4374	−0.107	0.368	0.284	0.341

Table S7. Sensitivity of the primary log-OLS model $\log(y + c) \sim T_{\text{chill},t-1} + T_{\text{CREC},t}$ to the offset constant c, using the full eight-year sample. The chill slope varies monotonically with c (from −0.961 at $c = 0.1$ to −0.697 at $c = 1.0$) but remains significant throughout ($p \leq 0.035$); sign and significance of the chill effect are preserved across the practical range of offsets.

c	n	$\beta_{T_{\text{chill}}}$ **(SE)**	p_{chill}	$\beta_{T_{\text{CREC}}}$	R^2_{log}	R^2_{raw}	$\textbf{RMSE}_{\text{raw}}$
0.10	8	−0.961 (0.306)	0.0255	+0.118	0.671	0.011	3.59
0.25	8	−0.899 (0.291)	0.0272	+0.105	0.663	0.053	3.51
0.50	8	−0.816 (0.272)	0.0299	+0.089	0.653	0.108	3.41
0.75	8	−0.751 (0.256)	0.0324	+0.077	0.643	0.149	3.33
1.00	8	−0.697 (0.242)	0.0347	+0.067	0.635	0.181	3.26

Table S8. Annual climate summary for the eight-year calibration sample (2016–2024, ex 2021). For each harvest year Y: parcel-mean olive yield with between-parcel SD ($n = 11$ parcels), chill-window ($Y - 1$) Dynamic-Model Chill Portions (CP) and Chill Hours below 12 °C (CH_{12}), chill-window mean temperature $T_{\text{chill},t-1}$, flowering-window and fruit-growth temperatures, fruit-growth ET_0 (FAO 56), and NOAA ONI averaged over the chill window. El Niño failure years (2016, 2024) in bold; triple-La Niña bumper year (2022) in italic.

Year Y	**Yield** (t ha^{-1})	**SD** (t ha^{-1})	**CP** (portions)	$\textbf{CH}_{12}$ (h)	$T_{\textbf{chill}}$ (°C)	$T_{\text{max},\textbf{FLOR}}$ (°C)	$T_{\textbf{mean},\textbf{FLOR}}$ (°C)	$T_{\text{max},\textbf{CREC}}$ (°C)	$T_{\textbf{CREC}}$ (°C)	$\textbf{ET}_{0,\textbf{CREC}}$ (mm)	$\textbf{ONI}_{\textbf{chill}}$ (°C)
2016	**1.60**	0.19	0.00	63	**18.20**	22.23	18.71	27.50	23.39	365.0	+1.44
2017	6.14	0.60	0.00	132	17.19	22.56	18.98	28.17	23.99	379.0	−0.08
2018	7.66	0.68	0.00	85	17.46	21.12	17.47	26.98	21.96	268.1	+0.23
2019	9.32	0.92	0.00	100	16.77	22.74	18.86	27.52	22.46	281.8	+0.09
2020	8.33	0.86	1.00	147	16.32	21.79	17.03	27.36	21.11	315.0	+0.42
2022	*12.06*	*1.08*	*3.03*	*417*	*16.56*	*22.09*	*15.95*	*24.77*	*18.66*	*247.6*	−0.37
2023	4.84	0.62	4.02	489	16.03	22.09	16.20	26.01	19.83	202.5	−0.83
2024	**0.68**	0.04	0.00	30	**18.45**	23.81	17.92	26.84	20.56	254.8	+0.98
Mean	6.33	—	1.01	183	17.25	22.30	17.64	26.90	21.50	289.2	+0.11